SRM Institute of Science and Technology College of Engineering and Technology

School of Computing

SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamil Nadu

Academic Year: 2024-25 (Even)

Test: CLAT-2 (Online Assessment) **Date: 21-04-2025 to 22-04-2025 (2 days)**

Course Code & Title: 21CSS301T – FULL STACK DEVELOPMENT **Duration**: 2 days

Year & Sem: III Year / VI Sem **Max. Marks:** 50

Course Articulation Matrix:

	PO												PSO		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	2	3
CO1	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**1**
CO3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2

1. **Group Task Name:** Student Team Members Management Application

2. **Technologies and Software to be Used:**

   - Frontend (**React.js**)
   - Backend (**Node.js + Express + MongoDB**)
   - **Visual Studio Code** – Code editor for writing and managing frontend/backend code
   - **MongoDB Compass** – GUI for managing MongoDB data
   - **Node.js** – Backend JavaScript run time

3. **App Structure – Pages & Styling:**

You have to create the following pages in your **React** application:

- Home Page
- Add Member Page
- View Members Page
- Member Details Page

You want to call (render) all these pages from your **App.js** using React Router. Also, apply **basic CSS styling** so the pages have a consistent look.

4. Functional Requirements:

1. Display Team Name
   - Show the team's name on the homepage or header.
2. Add New Team Members
   - Provide a form to input member details (e.g., name, role, contact info).
   - Submit form data to the backend to store in MongoDB.
3. View All Team Members
   - Fetch and display a list of all team members from the backend.
   - Use cards or a table for a clean layout.

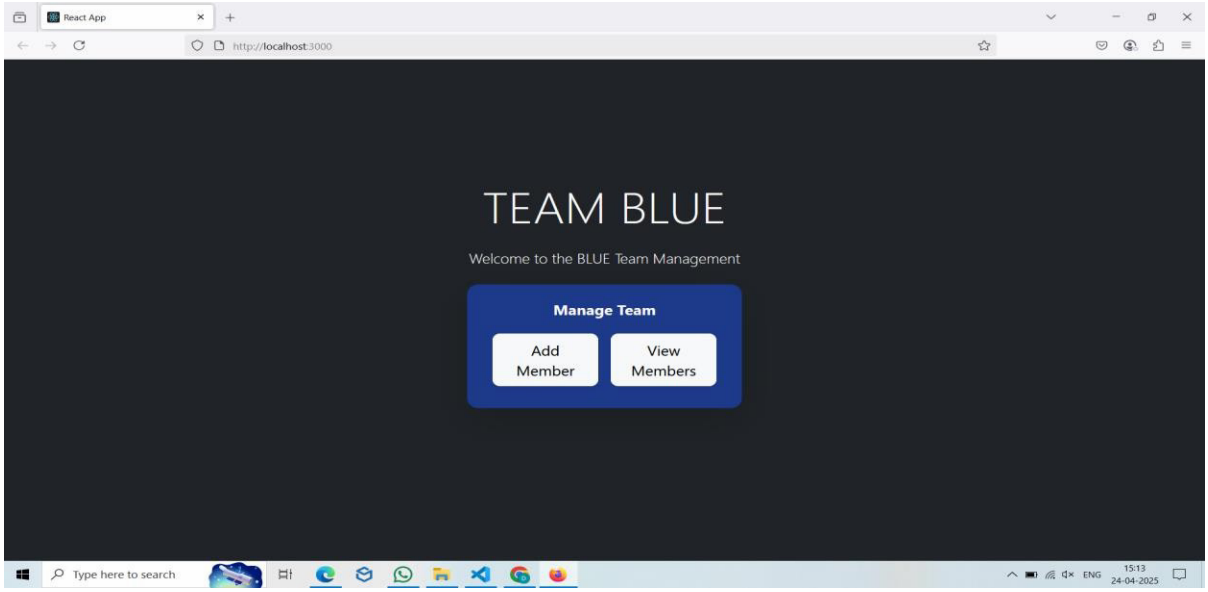
4. View Individual Member Details
    - On clicking a member, navigate to a detail page.
    - Fetch and display full details of the selected member.

5. Sample Expected output

1. Create a simple **landing page** that serves as the entry point of the app. **(10 Marks)**

   It should include:
   - A welcome message or team introduction.
   - Navigation buttons/links with the following functionality:
   - "Add Member" – Navigates to the form page where users can add a new team member.
   - "View Members" – Navigates to the page displaying a list of all team members.

   Design Tips:
   - Use a clean layout with centered content.
   - Include a header or banner with the team name.
   - Style buttons clearly to show they are clickable.

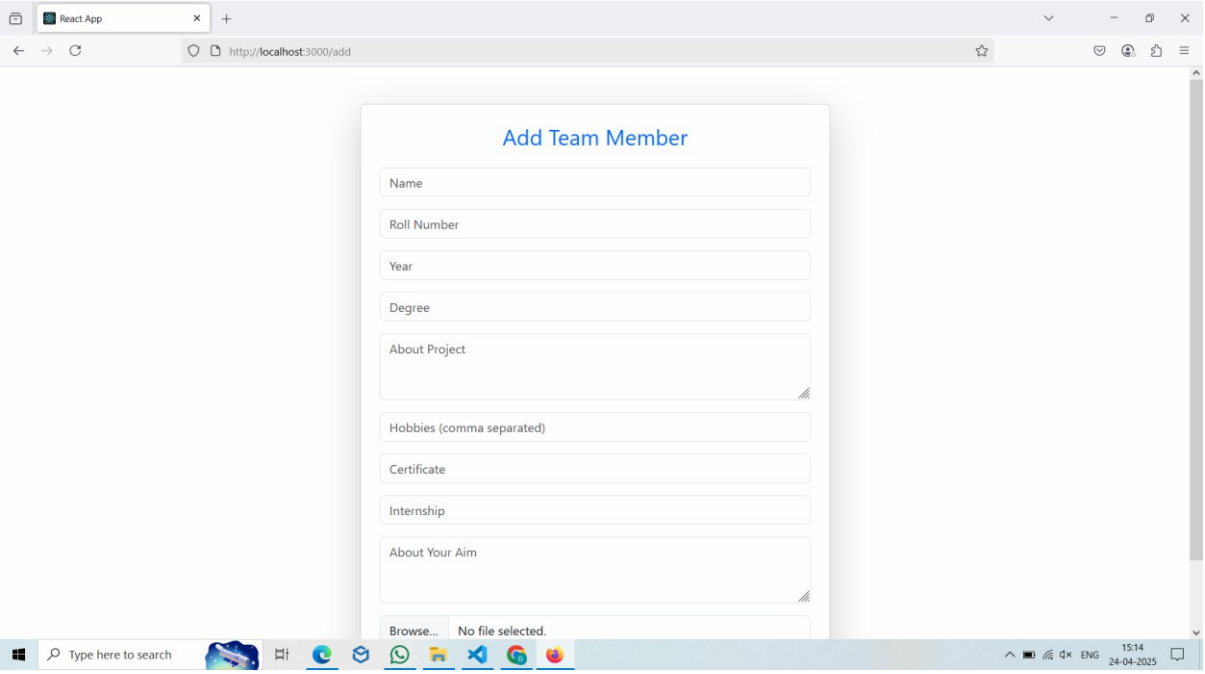


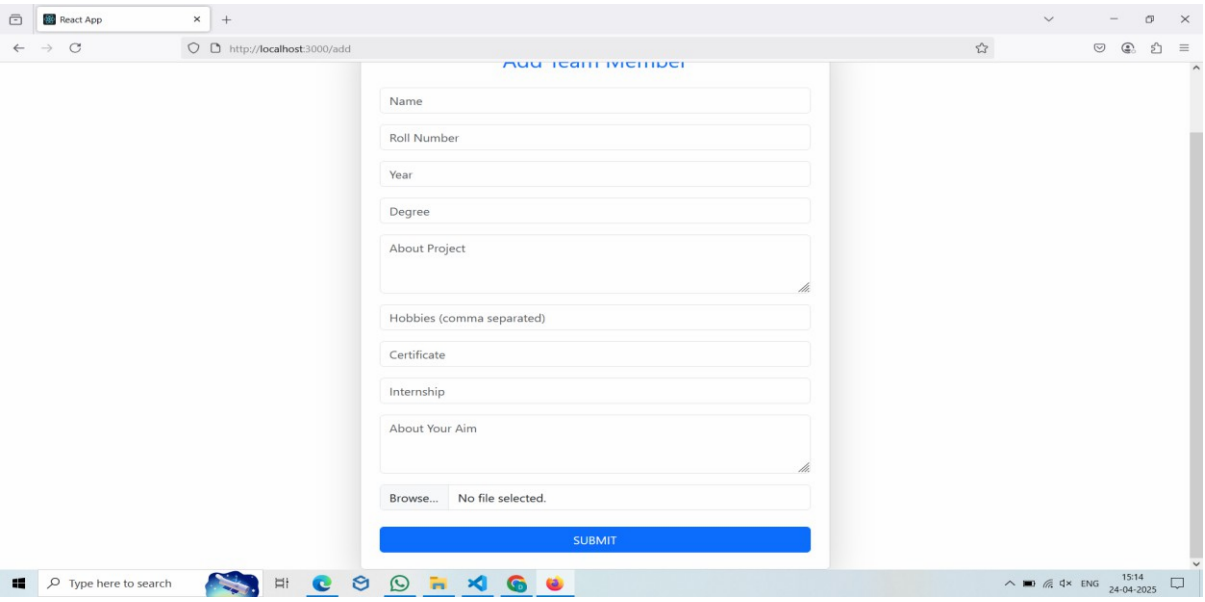
2. **Add Member Page** Instruction (With Image upload) **(10 Marks)**

   - Allow users to add a new team member by filling out a form.
   - Display a form with input fields for member details (e.g., name, role, email, etc.).
   - File input: **Upload image (can be your Professional photo)**
   - On form submission:
     - Validate input fields.
     - Send form data to the backend using Axios POST request.
     - The backend stores the data in MongoDB.
     - Save uploaded image to the uploads/ folder.

   API Used:

   POST /members



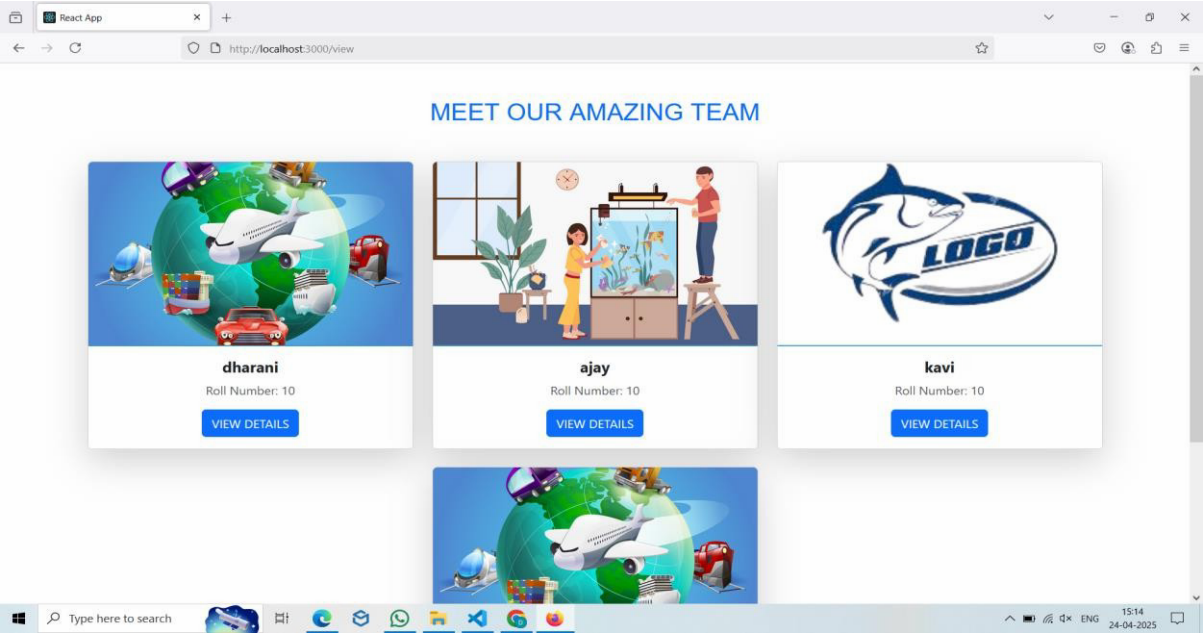


3. **View Members Page** Instruction **(5 Marks)**

- Display a list of all team members stored in the database.
- On page load:
  - Fetch all members from the backend using Axios GET request.
- Display each member's:
  - Name
  - Role
  - Profile image (from the uploads/ folder using the stored filename)
  - Include a “View Details” button/link

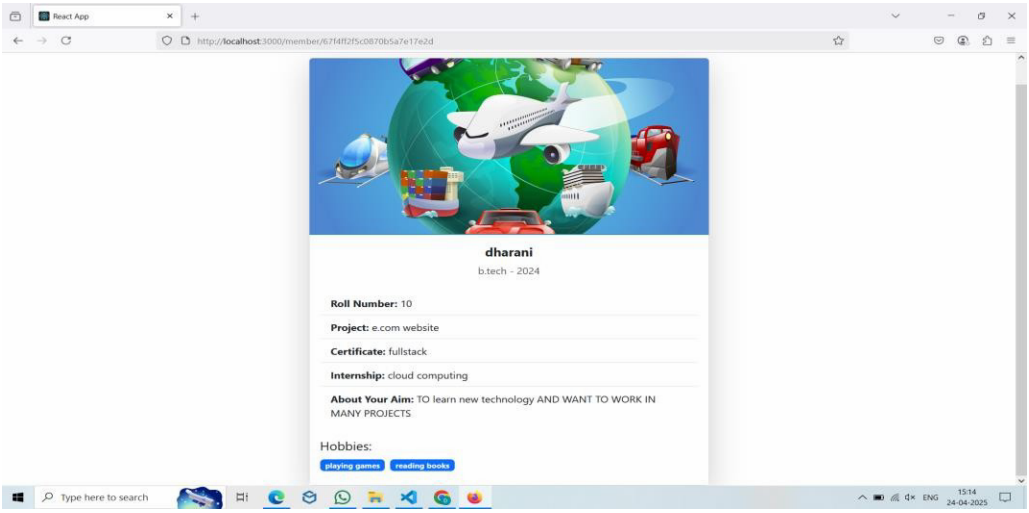
Use API:

GET /members



4. **Member Details Page** Instruction **(5 Marks)**

- Display full information about a single team member based on their unique ID. **When the page is accessed (e.g., via clicking "View Details" on the Members List page):**
- Use Axios GET request to fetch the specific member's details from the backend.
- Display the following:
  - Full name
  - Role
  - Email/contact
  - Profile image (from uploads/ folder)
  - Any additional member details
- Use API:
  GET /members/:id



5. **API Calls in Browser:** **(10 Marks)**

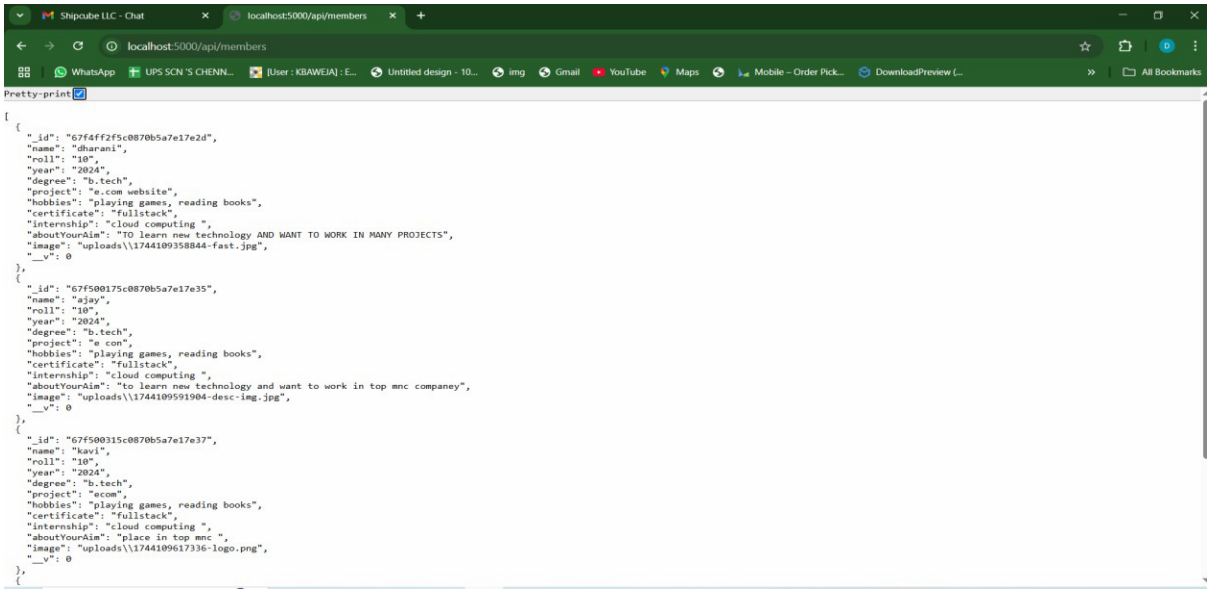
- These API endpoints facilitate data exchange between the frontend and backend. You can test them using a browser (for GET requests)

○GET /api/members

- Retrieves all team members
- Testable directly in the browser

○GET /api/members/:id

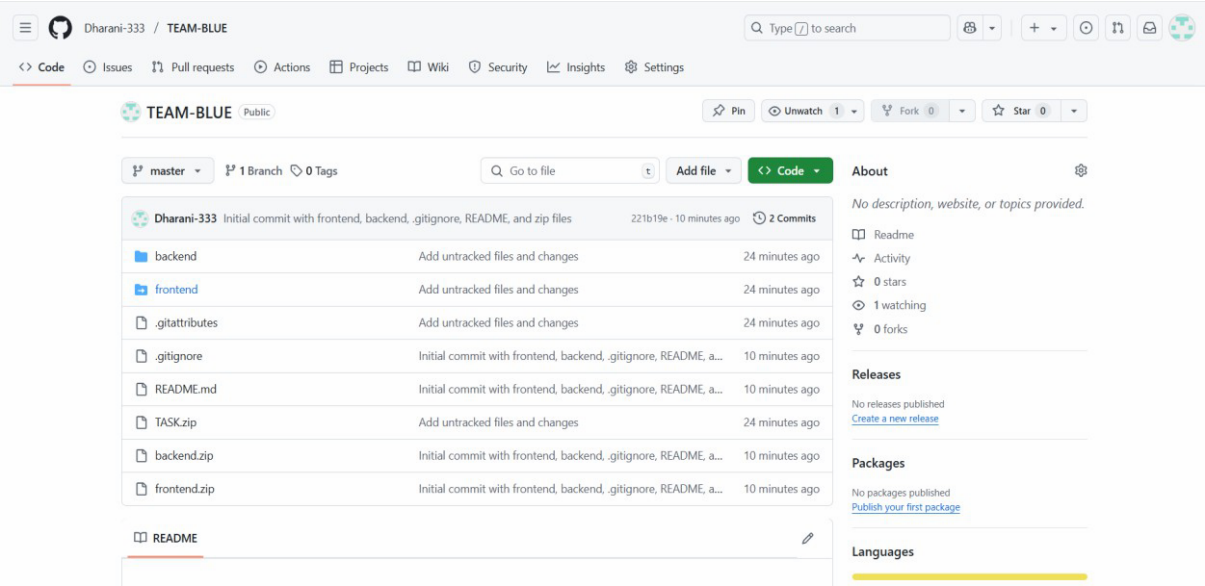
- Fetch details of a single member using their ID
- Replace: id with the actual member ID



6. Files to GitHub – Submission Instructions (10 Marks)

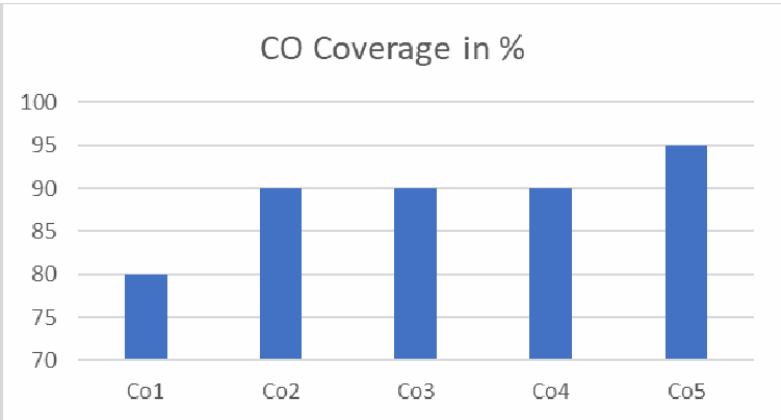
Submit your project by uploading all files to a GitHub repository named after your team name

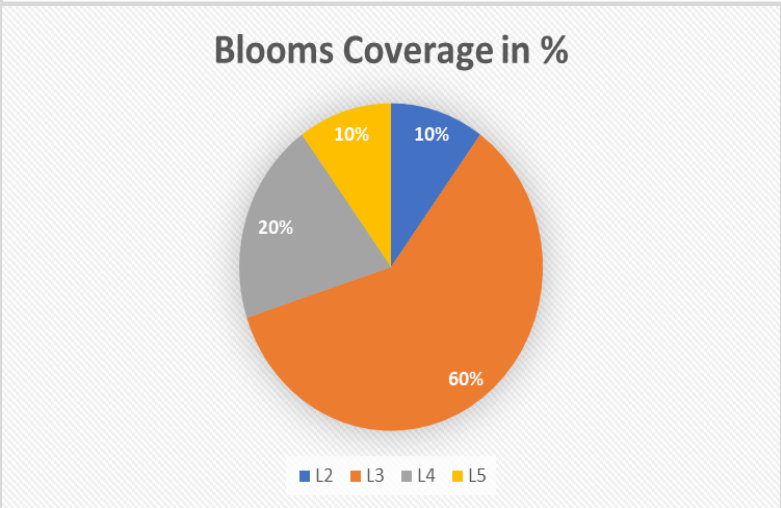
- Repository name should be your team name.
- Make it public
- Ensure all files and folders are properly structured as shown in the screen shot below
- Your submission will be **evaluated based on the time** you pushed the code to GitHub.
- Add a. gitignore file to exclude `node_modules`, etc.
- Include a clear **README.md** with:
  - Project description
  - Installation steps
  - API endpoints
  - How to run the app



Note: Submit your GitHub repository link through the Google Form that was shared before the deadline (1-5-2025, 12:00pm)

Course Outcome (CO) and Bloom's level (BL) Coverage in this Assessment :





Approved by the Audit Professor/Course Coordinator